[INCH-POUND]

MIL-C-39001A
AMENDMENT 8
27 October 1989
SUPERSEDING
AMENDMENT 7
22 February 1985

MILITARY SPECIFICATION

CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-C-39001A, dated 6 April 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Following title, add:

"

| |
|---|
| FAILURE RATE LEVEL 'L' IS INACTIVE |
| FOR DESIGN AFTER 29 July 1983 |

"

1.2.1: In type designation, delete "010" and substitute "1R0".

1.2.1.3, delete and substitute:

"1.2.1.3 Capacitance. The nominal capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow. When the nominal value is less than 10 pF, the letter 'R' shall be used to indicate the decimal point and the succeeding digit(s) of the group shall represent significant figure(s). For example 1R0 indicates 1.0 pF; R75 indicates 0.75 pF; and 0R5 indicates 0.5 pF."

PAGE 2

Table V, delete symbol "T" (and accompanying failure rate percentage). Make this same deletion wherever "T" appears in the specification.

PAGE 3

2.1, list of military specifications, add:

"MIL-C-5 - Capacitors, Fixed, Mica Dielectric, General Specification For."

3.2, line 2: At end of paragraph, add "Unless procured from the manufacturer or his authorized distributor listed or approved for listing on the qualified products list, parts furnished under this specification shall not be considered as having met the requirements of this specification."



AMSC N/A FSC 5910

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PAGE 6

Figure 2, delete and substitute the following:

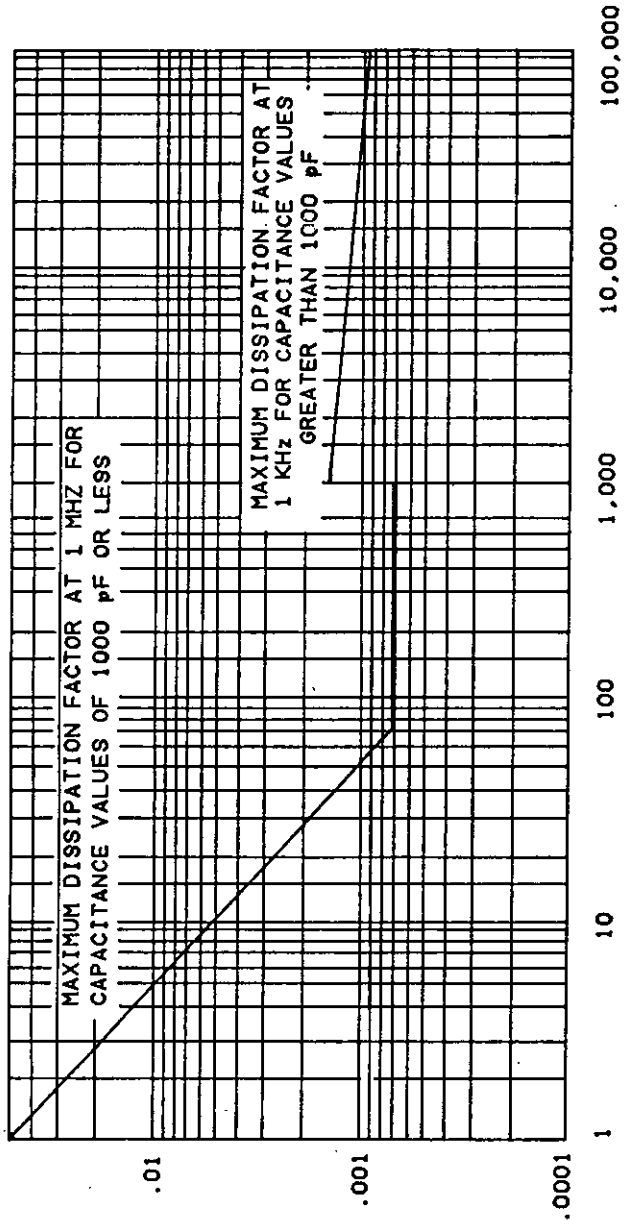



FIGURE 2. Capacitance vs dissipation factor.



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PAGE 7

3.15, title, delete "Temperature cycling" and substitute "Thermal shock". Make this same change wherever "Temperature cycling" appears in this specification.

PAGE 8

3.22: Following "Insulation resistance", delete "Shall not exceed initial requirement" and substitute "Shall not be less than the initial requirement".

Following 3.23.1, add:

"3.23.2 Nonestablished reliability (non-ER) marking. An established reliability (ER) part manufactured in accordance with this specification may be marked and furnished as a non-ER part from MIL-C-5, if produced on the same assembly line, and provided it is subjected to and meets all the inspection requirements of the ER part."

PAGE 9

4.1.1, delete and substitute:

"4.1.1 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790, with the following exceptions:

(a) Only the following of paragraph 5.2.7(j) shall apply: 'the manufacturer shall be as a minmum be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to a production lot.'

(b) Paragraph 5.2.11.3 shall not apply."

PAGE 10

4.4.4(c), line 1, delete "Maintenance period A" and substitute "Maintenance period B".

PAGE 12

4.6.1.1, delete and substitute:

"4.6.1.1 Inspection lot. An inspection lot shall be as defined in MIL-STD-105 and shall consist of capacitors in one or more styles, produced during a maximum period of 1 month. The sample selected from the lot shall be representative of the case sizes and styles in the lot, with the unit selected for test having the highest capacitance value in each case size. Styles CMR03, CMR04, CMR06, CMR07, and CMR08 may be combined for quality conformance and FR level maintenance."

4.6.1.2.1, delete second and last sentences.



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* TABLE VIII, delete and substitute:

"TABLE VIII. Group A inspection.

| Examination or test | Requirement paragraph | Method paragraph | Number of samples |
|---|---|---|---|
| Subgroup 1 (FR levels M, P, R, and S) | | | |
| High voltage stabilization | 3.11 | 4.7.7 | (100% inspection) |
| Subgroup 2A (FR level L only) | | | |
| Dielectric withstanding voltage | 3.6 | 4.7.2 | 32 samples 0 failures |
| Insulation resistance | 3.8 | 4.7.4 | |
| Capacitance | 3.9 | 4.7.5 | |
| Dissipation factor | 3.10 | 4.7.6 | |
| Subgroup 2B (FR levels M, P, R, and S) | | | |
| Capacitance | 3.9 | 4.7.5 | (100% inspection) |
| Dissipation factor | 3.10 | 4.7.6 | |
| Subgroup 3 | | | |
| Visual and mechanical examination | 3.1, 3.4 to 3.5.2 incl, 3.23 to 3.25.1 incl | 4.7.1 | FR level L 8 samples 0 failures<br>FR level M,P,R,S. 13 samples 0 failures |

"

PAGE 14

4.6.2.1.1.1, line 2, delete "every month" and substitute "every 2 months".

PAGE 15

TABLE X, subgroup 2, following "Moisture resistance", insert "2/". Insert the following at end of table:

"2/ For FR level L, 10 cycles are required."

PAGE 17

4.7.7, delete last sentence and substitute "Insulation resistance shall then be measured as specified in 4.7.4.1."

PAGE 18

4.7.11.1, line 1, delete "method 102" and substitute "method 107".

4.7.11.1(a), line 1, delete "Test condition - D" and substitute "Test condition - A".



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4.7.14(d), delete and substitute:

"(d) Polarization - A dc potential of 100 volts or rated volts, whichever is less, shall be applied across the capacitor terminals of 50 percent of the units under test and no voltage applied to the other 50 percent."

* "4.7.15.1.1, delete the first sentence and substitute "Rated conditions. Capacitors shall be subjected to a minimum of 100 percent of rated dc voltage at +125°C ± 3°C or +150°C ± 3°C, as applicable, for 2,000 +72 -0 hours."

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

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Preparing activity:
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DLA - ES

(Project 5910-1658)



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